You Know DSA.

But Can You Solve a NEW Problem? 🤔

Learning DSA ≠ Problem Solving

Knowledge is the starting point.
Problem solving is knowing what to do next.

NEW
PROBLEM

The Common Situation

You know the data structures. But when a new problem appears, where do you start?

You’ve learned all this…

- Arrays
- Stack
- Trees
- Binary Search
- HashMap
- Queue
- Graphs

New Problem → **Where do I start?**

What’s Missing?

Knowing WHAT isn’t enough.

You also need to know:

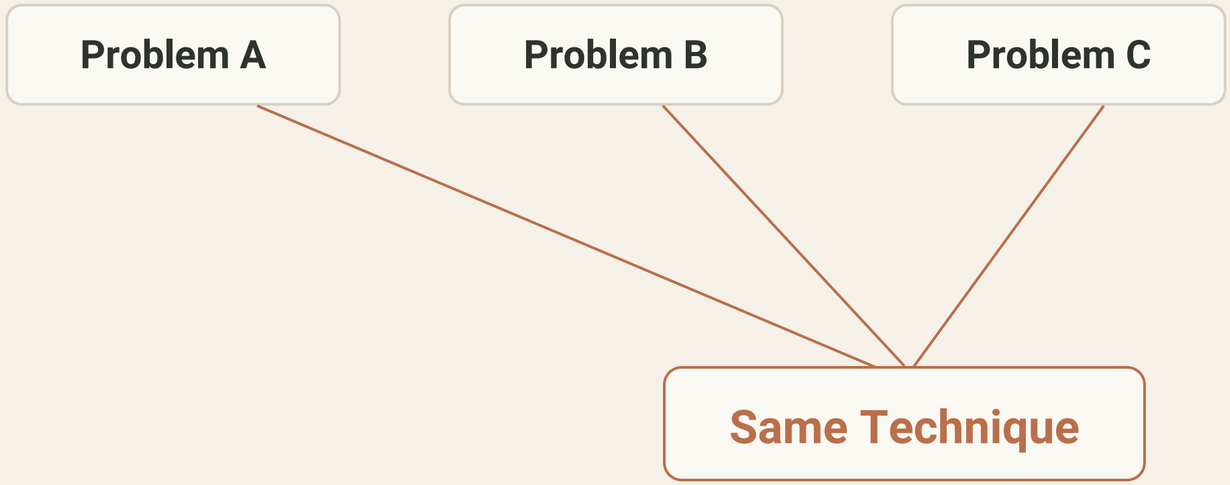
WHEN should I use it?

That’s where Patterns help.



What is a DSA Pattern?

Different Problems. Similar Thinking.



Don’t memorize solutions.
Recognize approaches.

Let’s Try One

Two Pointers

Given a sorted array…

[1, 2, 4, 6, 8, 9]

Target = 10

Find a pair whose sum = 10

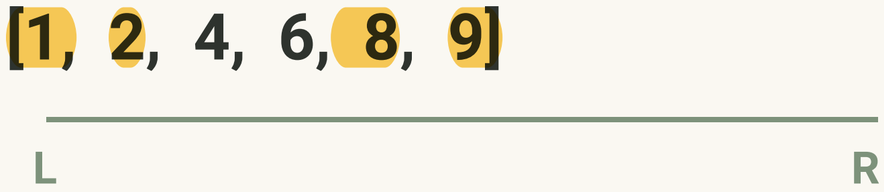
First thought?

Brute Force → O(n²)

Look for Clues 👀

What can we take advantage of?

- ✓ Sorted
- ✓ Looking for a pair
- ✓ Target comparison
- ✓ Can we beat O(n²)?



Two Pointers

Another Problem

Sliding Window

Maximum sum of 3 consecutive elements

[2, 1, 5, 1, 3, 2]

[2, 1, 5] [1, 5, 1] [5, 1, 3]

Clues:

- ✓ Consecutive elements
- ✓ Fixed size
- ✓ Repeated calculation

Sliding Window

Clues ≠ Rules

Patterns should guide your thinking,
not replace it.

What This Series Will Teach

One Pattern at a Time.

Understand

↓

Identify Clues

↓

Brute Force

↓

Optimize

↓

Practice Variations

The goal

Understand the idea.
Spot the clues.
Choose the right approach.

Next → Two Pointers

Practice,
Don’t Just Watch

Learn → Recognize → **Practice**

DSA Patterns
codemeal.in/patterns

DSA Roadmaps
codemeal.in/roadmaps

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